

Name of Student: _____

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

dr. rahul chaurasiya Mid-Term Examination September - 2025

Subject Name/ Code: Computer Organization and Architecture (COA)/ CSE-3003 MM:20
Time Allotted: 90 min

NOTE: 1. All questions are compulsory.
2. Mark answer number clearly. Answers without precision receive NO credit.

1. A	Compare different cache mapping techniques (Direct-Mapped, Fully Associative, and Set-Associative) in terms of conflict misses when mapping data blocks from main memory to cache. Analyze and Compute Best and Worst-case probability of conflict misses for each technique. Based on your analysis, suggest the most efficient technique to minimize conflict misses while considering practical implementation constraints about main memory and cache memory size.	3
✓ B.	Explain the statement that pipelining is used to reduce the average instruction processing time rather than the execution time of a single instruction.	2
✓ 2. A	Consider a cache memory with 80% Hit for Read operation and 90% HIT for write operation. The cache access time is 5 times faster than main memory access time, which is 20ms. If Write-back updation is used then if there is a Miss for read or write, a 2-word block (1 block =1 word) must be moved from main memory to cache memory. Assume that cache is initially full and at any point of time 20% of its block have modified. Compute Avg. Read access time and Avg. write access time.	3
✓ B.	Discuss the consequences of Cache Coherence problem and provide a comparative analysis of the techniques used to address the issue.	2
✓ 3. A	Consider a hypothetical computer that supports one-address, two-address, and three-address instruction formats. Each instruction is 24 bits long and is stored in a 128-word main memory. If the system is using 2 two-address instructions and 3 three-address instructions, how many one-address instructions can the system support? Consider, no other opcodes are reserved.	3
✓ B.	Consider an initially empty cache containing <u>16</u> blocks. The MM has 256 blocks. The following MM blocks are referred following LRU replacement	2

P.T.O.

we modules with

$$S = \frac{\text{Blocks}}{K} = \frac{16}{2} = 8$$

$$\frac{79}{32} = 2.46875$$

$$\frac{159}{8} = 19.875$$

	policy, if needed. 0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155. At the end of references, what will be status of the cache with two-way set associate Mapping.	
4. A.	Discuss how operand forwarding technique is useful in reducing Stall Penalty in Pipeline processing. Then consider a five-stage instruction Pipeline with instruction fetch, instruction decode, operand fetch, perform operation and write operand stages. All the stages except perform operation (PO) consume 1-clock/instruction/stage. The perform operation requires 6 clocks for Division, 3 Clocks for Multiplication and 2 clocks each for Addition and Subtraction. How many minimum clocks are required to implement the following four instructions with operand forwarding. I0: MUL R2 R0 R1 ✓ I1: DIV R5 R3 R4 ✓ I2: ADD R2 R5 R2 I3: SUB R5 R2 R6	3
B.	If it is required to implement a 4-Stage Pipeline such that Stage-2 delay is twice the Stage-1 delay and half of Stage-3 delay. Stage-4 and Stage-2 delay are same. The Stage-3 delay is 40ns, then how much time is required to complete 100 instructions and what efficiency can be achieved with this Pipeline?	2

without = 100 * all cycle time

All the Best

3(A)

$$2^{24} = 2 \cdot 2^{14} + 2^{21} \cdot 3 + x \cdot 2^7$$

$$2^{24} = 2^{15} + 2^{21} \cdot 3 + x \cdot 2^7$$

$$16777216 = 32768 + 6291456 + x \cdot 2^7$$

$$81664$$